## *DATA EXPLORE*

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
#import all the library
```

```
df=pd.read_csv('/content/crop_yield.csv')
#import required dataset
```

```
df.head(10)
#to check 1st 10 rows
```

| | Crop | Crop_Year | Season | State | Area | Production | Annual_Rainfall | Fe |
|---|---|---|---|---|---|---|---|---|
| **0** | Arecanut | 1997 | Whole Year | Assam | 73814.0 | 56708 | 2051.4 | 70 |
| **1** | Arhar/Tur | 1997 | Kharif | Assam | 6637.0 | 4685 | 2051.4 | 6 |
| **2** | Castor seed | 1997 | Kharif | Assam | 796.0 | 22 | 2051.4 | |
| **3** | Coconut | 1997 | Whole Year | Assam | 19656.0 | 126905000 | 2051.4 | 18 |
| **4** | Cotton(lint) | 1997 | Kharif | Assam | 1739.0 | 794 | 2051.4 | 1 |
| **5** | Dry chillies | 1997 | Whole Year | Assam | 13587.0 | 9073 | 2051.4 | 12 |
| **6** | Gram | 1997 | Rabi | Assam | 2979.0 | 1507 | 2051.4 | 2 |
| **7** | Jute | 1997 | Kharif | Assam | 94520.0 | 904095 | 2051.4 | 89 |
| **8** | Linseed | 1997 | Rabi | Assam | 10098.0 | 5158 | 2051.4 | 9 |
| **9** | Maize | 1997 | Kharif | Assam | 19216.0 | 14721 | 2051.4 | 18 |

```
df.shape
#to check number of column and rows present in the dataset
```

```
(19689, 10)
```

```
df.columns
#it returns all the columns name present in the data set
```

```
Index(['Crop', 'Crop_Year', 'Season', 'State', 'Area', 'Production',
       'Annual_Rainfall', 'Fertilizer', 'Pesticide', 'Yield'],
      dtype='object')
```

```
df.describe()
#it provide all the statoistical value of numerical columns
```

| | Crop_Year | Area | Production | Annual_Rainfall | Fertilizer | |
|---|---|---|---|---|---|---|
| **count** | 19689.000000 | 1.968900e+04 | 1.968900e+04 | 19689.000000 | 1.968900e+04 | 1 |
| **mean** | 2009.127584 | 1.799266e+05 | 1.643594e+07 | 1437.755177 | 2.410331e+07 | 4 |
| **std** | 6.498099 | 7.328287e+05 | 2.630568e+08 | 816.909589 | 9.494600e+07 | 2 |
| **min** | 1997.000000 | 5.000000e-01 | 0.000000e+00 | 301.300000 | 5.417000e+01 | 9 |
| **25%** | 2004.000000 | 1.390000e+03 | 1.393000e+03 | 940.700000 | 1.880146e+05 | 3 |
| **50%** | 2010.000000 | 9.317000e+03 | 1.380400e+04 | 1247.600000 | 1.234957e+06 | 2 |
| **75%** | 2015.000000 | 7.511200e+04 | 1.227180e+05 | 1643.700000 | 1.000385e+07 | 2 |
| **max** | 2020.000000 | 5.080810e+07 | 6.326000e+09 | 6552.700000 | 4.835407e+09 | 1 |

```
df.info()
#it provide quick summary of a DataFrame
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 19689 entries, 0 to 19688
Data columns (total 10 columns):
 #   Column           Non-Null Count  Dtype  
---  ------           --------------  -----  
 0   Crop             19689 non-null  object 
 1   Crop_Year        19689 non-null  int64  
 2   Season           19689 non-null  object 
 3   State            19689 non-null  object 
 4   Area             19689 non-null  float64
 5   Production       19689 non-null  int64  
 6   Annual_Rainfall  19689 non-null  float64
 7   Fertilizer       19689 non-null  float64
 8   Pesticide        19689 non-null  float64
 9   Yield            19689 non-null  float64
dtypes: float64(5), int64(2), object(3)
memory usage: 1.5+ MB
```

```
inti=df.select_dtypes(include='int64').columns
#it returns all integer typoe data
```

```
flo=df.select_dtypes(include='float64').columns
#it return all float type data
```

```
cat=df.select_dtypes(include='object').columns
#it returns all object types column
```

```
num=df.select_dtypes(include=['int64','float64']).columns
```

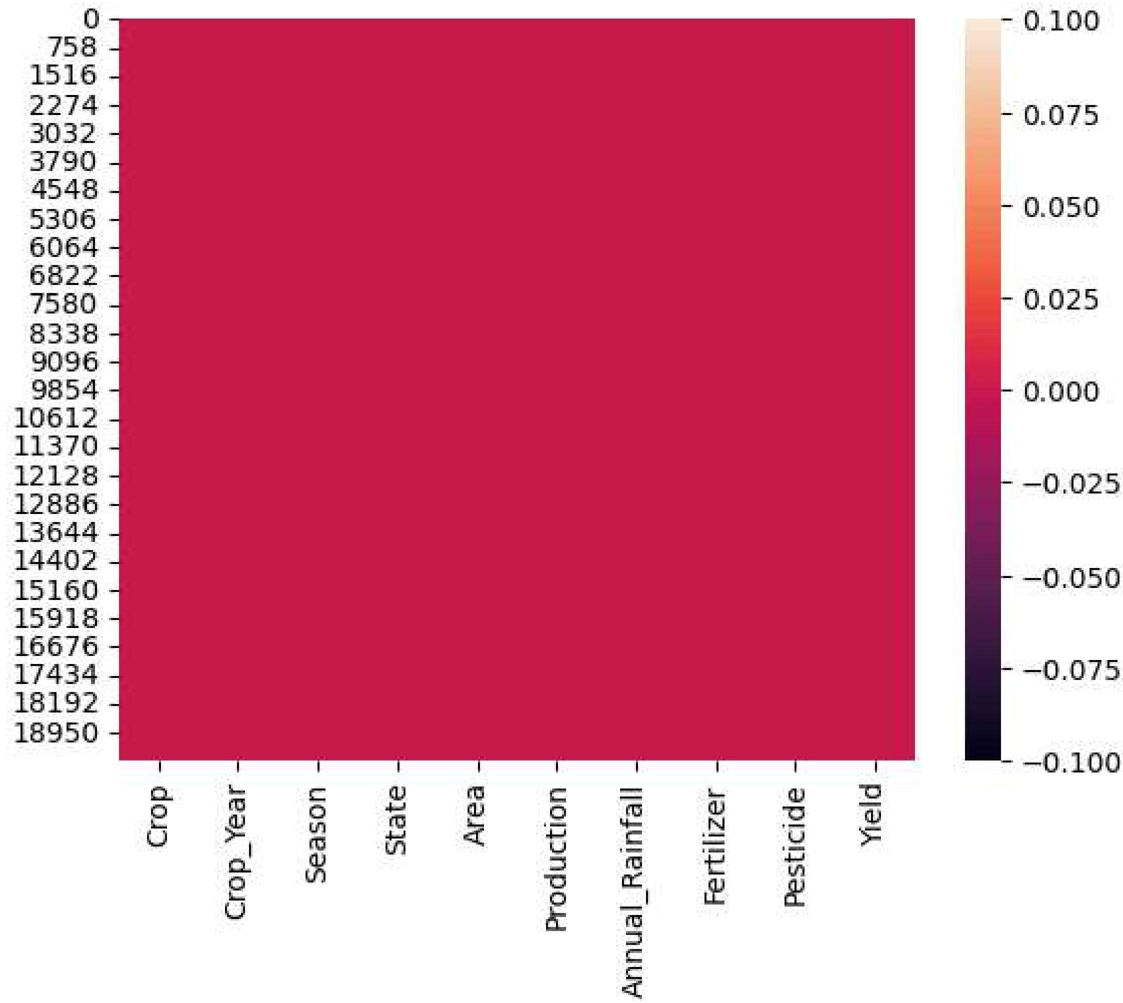
```
df.isnull().sum()
#it is used to check there is any missing value present in any column
```

| | 0 |
|---|---|
| **Crop** | 0 |
| **Crop_Year** | 0 |
| **Season** | 0 |
| **State** | 0 |
| **Area** | 0 |
| **Production** | 0 |
| **Annual_Rainfall** | 0 |
| **Fertilizer** | 0 |
| **Pesticide** | 0 |
| **Yield** | 0 |

**dtype:** int64

```
sns.heatmap(df.isnull())
```

```
<Axes: >
```




```
df.duplicated().sum()
```

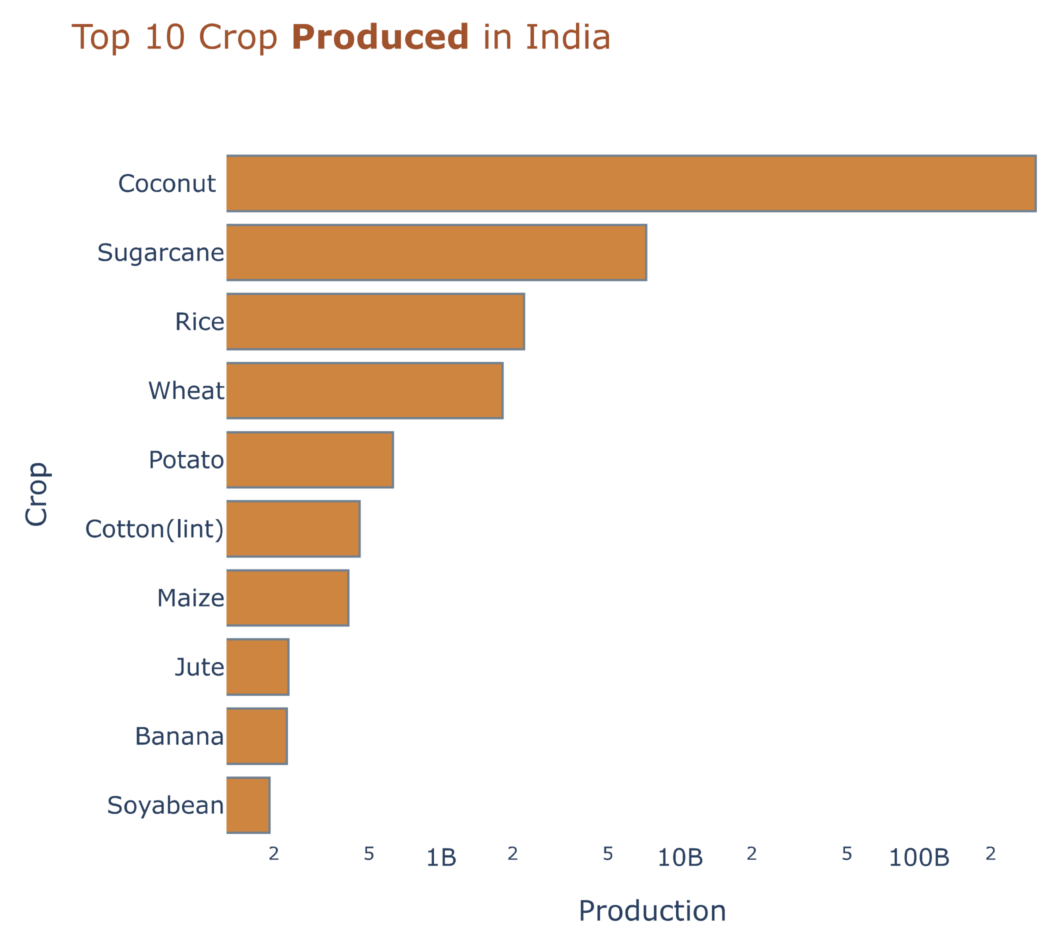
```
np.int64(0)
```

EXPLANATORY DATA ANALYSIS

```
crop_p=df.groupby('Crop')[['Production']].sum()
crop_production=crop_p.sort_values(by=['Production'],ascending=False).head(10)
crop_production
```

| | Production |
|---|---|
| **Crop** | |
| **Coconut** | 308751970278 |
| **Sugarcane** | 7236536755 |
| **Rice** | 2229301180 |
| **Wheat** | 1810754504 |
| **Potato** | 629688035 |
| **Cotton(lint)** | 457831414 |
| **Maize** | 410911181 |
| **Jute** | 230423821 |
| **Banana** | 226471979 |
| **Soyabean** | 191659664 |

```
from plotly import plot
fig=px.bar(crop_production,x='Production',y=crop_production.index,log_x=True,ti
fig.update_yaxes(autorange="reversed")
fig.update_layout(xaxis_title='Production',yaxis_title='Crop')
fig.update_traces(marker_color='Peru',marker_line_color='slategray',marker_line
fig.update_layout(title_font_color='sienna',plot_bgcolor='white')
fig.show()
```

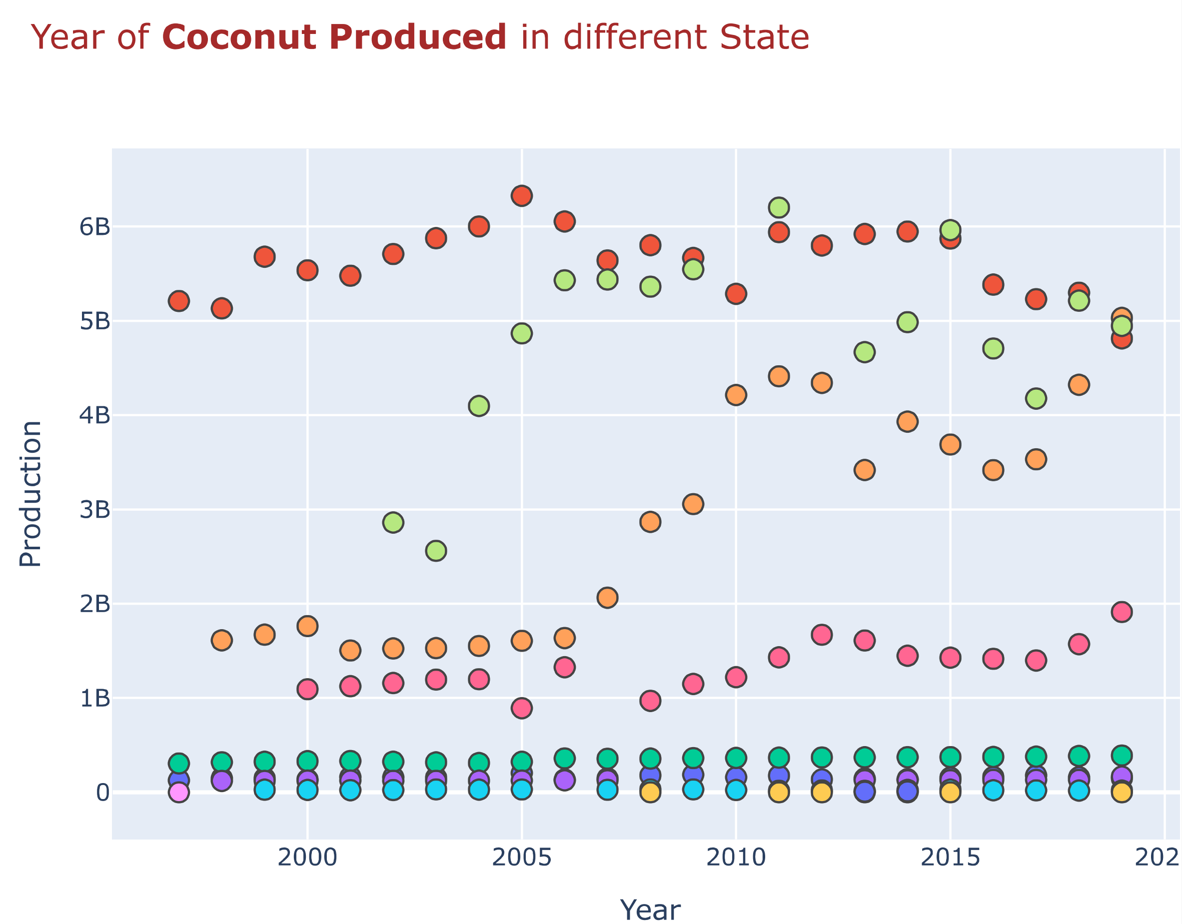



The graph indicates the top 10 crops produced in India. Coconut, sugarcane and rice are the top three produced crops, where production of coconut surpasses other crops with a huge amount of 308.752B. While the production of sugarcane and rice are 7.236B and 2.229B followed by wheat production which accounts to 1.811B.

```
coco_p = df[df["Crop"] == 'Coconut ']
coco_p.head()
```

| | Crop | Crop_Year | Season | State | Area | Production | Annual_Rainfall | F |
|---|---|---|---|---|---|---|---|---|
| **3** | Coconut | 1997 | Whole Year | Assam | 19656.0 | 126905000 | 2051.4 | |
| **60** | Coconut | 1997 | Whole Year | Kerala | 884344.0 | 5210000000 | 3252.4 | 8 |
| **94** | Coconut | 1997 | Whole Year | West Bengal | 24273.0 | 306202300 | 1852.9 | |
| **150** | Coconut | 1998 | Whole Year | Assam | 20166.0 | 149866000 | 2354.4 | |
| **178** | Coconut | 1998 | Whole Year | Goa | 24858.0 | 121000000 | 2964.9 | |

```
fig=px.scatter(data_frame=coco_p, x='Crop_Year',y='Production',color='State',ti
fig.update_xaxes(title='Year')
fig.update_traces(marker_line_width=1,marker_size=10)
fig.update_layout(title_font_color='Brown')
fig.show()
```



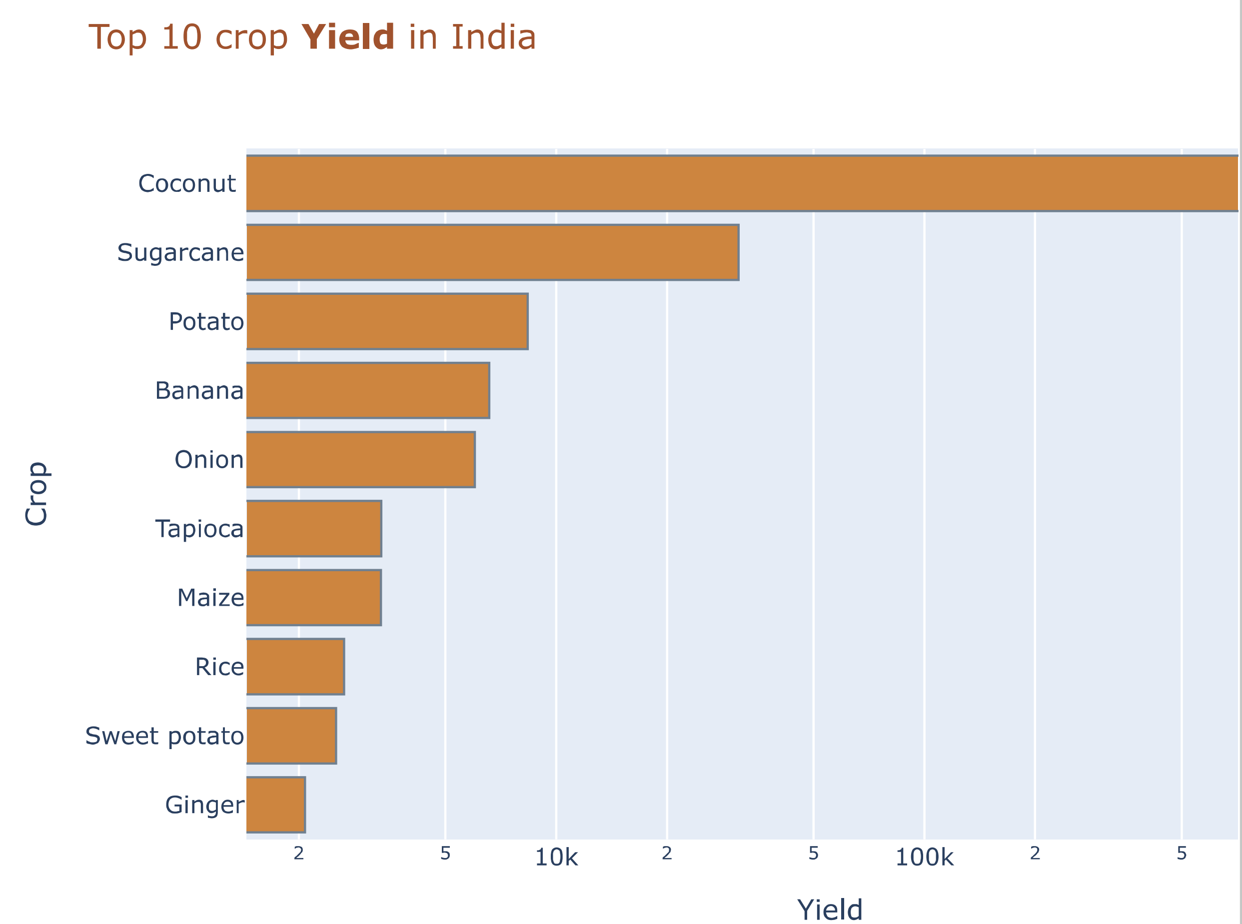

The graph indicates that Kerala is the highest coconut producing state followed by Karataka and Tamil Nadu from 1997 to 2019. Although there are some exceptions, in 2011 and 2015 Tamil Nadu surpassed Kerala in coconut production while in 2019 Karnataka was the highest coconut producing state followed by Tamil Nadu and Kerala. The graph also shows that in 1997 there was no production of coconut in Karnataka. In Tamil Nadu also there was no production of coconut before 2002, there was also no production in 2010 and 2012.

Production of coconut with slight increase in all the years is noticed in West Bengal and Assam while Kerala has increase and decrease in few years. Andhra Pradesh which is the fourth coconut producing state in India had no production before 2000, after which production increased in all the years except in 2007 when there was no production.

```
crop_y=df.groupby('Crop')[['Yield']].sum()
crop_yield=crop_y.sort_values(by=['Yield'],ascending=False).head(10)
crop_yield
```

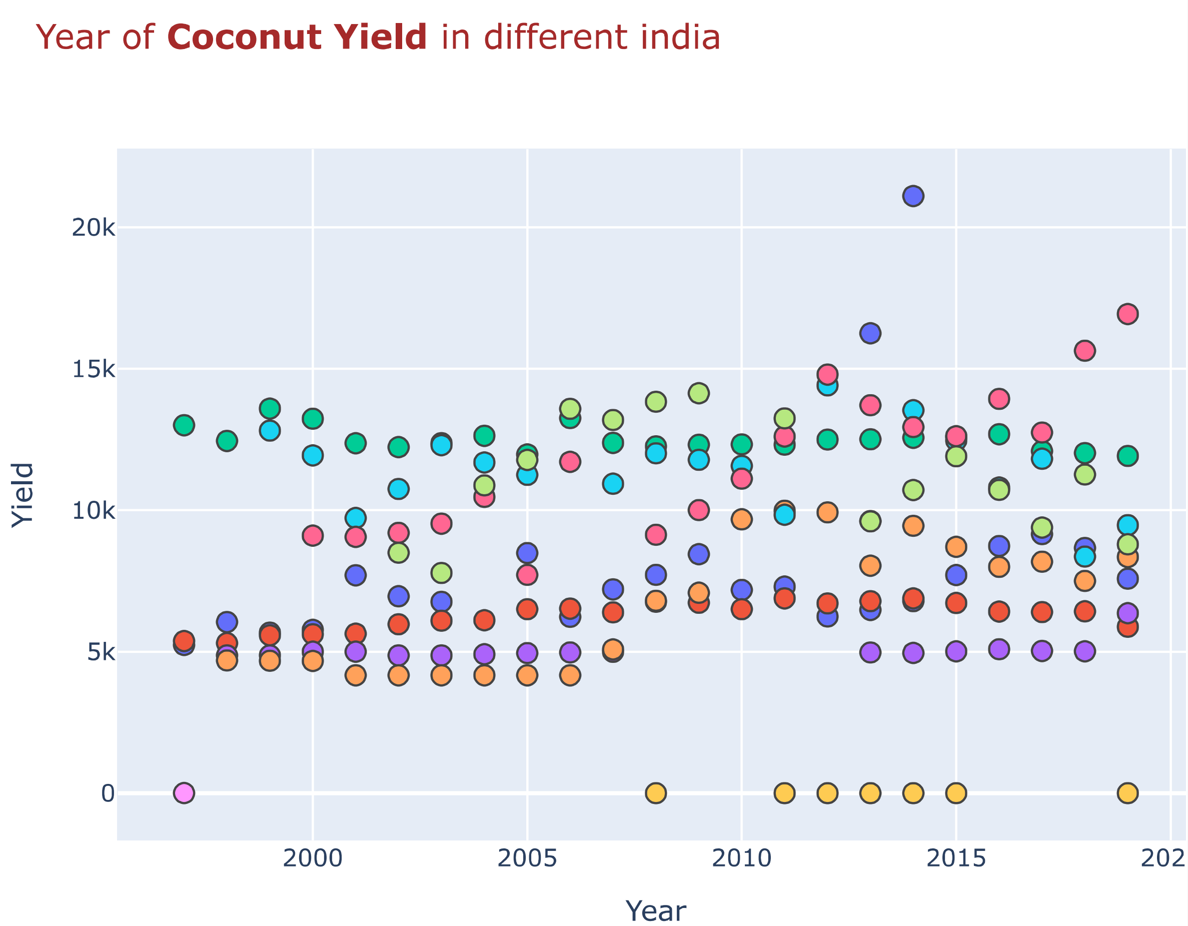
| Crop | Yield |
|---|---|
| Coconut | 1.488144e+06 |
| Sugarcane | 3.129510e+04 |
| Potato | 8.372319e+03 |
| Banana | 6.578526e+03 |
| Onion | 6.014376e+03 |
| Tapioca | 3.350128e+03 |
| Maize | 3.341536e+03 |
| Rice | 2.655538e+03 |
| Sweet potato | 2.522735e+03 |
| Ginger | 2.080831e+03 |

```
from plotly.graph_objs import Marker
fig=px.bar(crop_yield,x='Yield',y=crop_yield.index,title='Top 10 crop <b>Yield<
fig.update_yaxes(autorange='reversed')
fig.update_traces(marker_line_width=1,orientation='h',marker_color='Peru',marke
fig.update_layout(title_font_color='Sienna')
fig.show()
```




The graph indicates the yield of top 10 crops in India. Coconut, sugarcane and potato are the top three crops with highest yield, where yield of coconut surpasses other crops with a huge amount of 1.488M. While the yield of sugarcane and potato are 31.295k and 8372.319 followed by wheat production which accounts to 6578.526.

```
fig=px.scatter(coco_p,x='Crop_Year',y='Yield',color='State',title='Year of <b>C
fig.update_traces(marker_line_width=1,marker_size=10)
fig.update_xaxes(title='Year')
fig.update_layout(title_font_color='Brown')
fig.show()
```

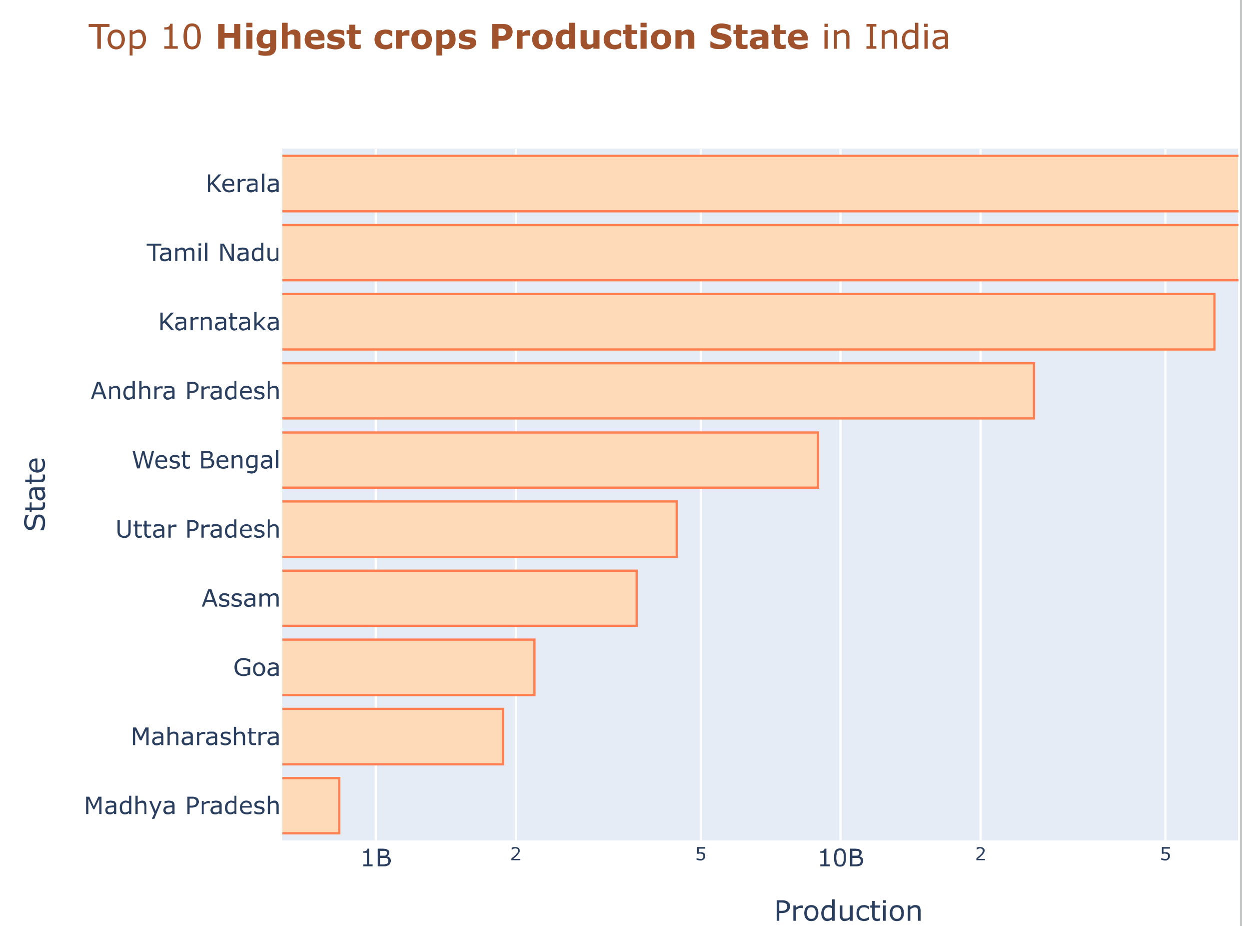



The graph indicates that yield of coconut is high in West Bengal, Puducherry, Tamil Nadu and Andhra Pradesh, where the highest yield varies among these states

```
state_p=df.groupby('State')[['Production']].sum()
state_production=state_p.sort_values(by=['Production'],ascending=False).head(1(
state_production
```

| State | Production |
|---|---|
| Kerala | 129700649853 |
| Tamil Nadu | 78051759253 |
| Karnataka | 63772797366 |
| Andhra Pradesh | 26076218605 |
| West Bengal | 8941179120 |
| Uttar Pradesh | 4442585302 |
| Assam | 3637714928 |
| Goa | 2193998349 |
| Maharashtra | 1878564915 |
| Madhya Pradesh | 834490323 |

```
fig=px.bar(state_production,x='Production',y=state_production.index,log_x=True,
fig.update_yaxes(autorange='reversed')
fig.update_traces(marker_color='peachpuff',marker_line_color='coral',orientatic
fig.update_layout(title_font_color='sienna')
fig.show()
```



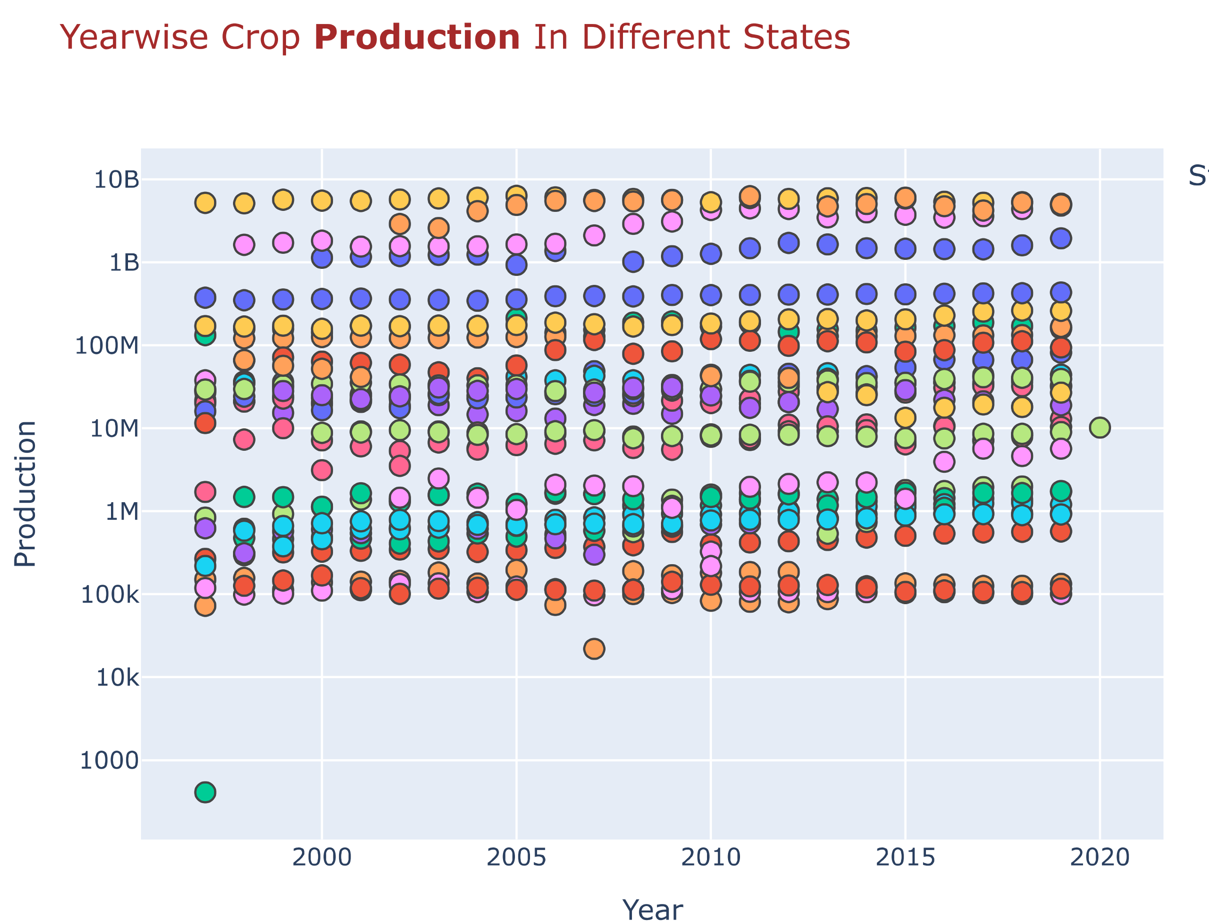

The graph indicates the top 10 crop producing states in India. Kerala, Tamil Nadu and Karnataka are the top three states with highest crop production, where production of crops in Kerala is 129.7006B. While in Tamil Nadu and Karnataka crop production are 78.051B and 63,772B followed by Andhra Pradesh where production is 26.076B.

```
#Crop production in different states
year = df.groupby(['Crop_Year','State'])[['Production']].sum()
crop_year = year.reset_index()
crop_year
```

| | Crop_Year | State | Production |
|---|---|---|---|
| **0** | 1997 | Andhra Pradesh | 28552600 |
| **1** | 1997 | Arunachal Pradesh | 267148 |
| **2** | 1997 | Assam | 132556429 |
| **3** | 1997 | Bihar | 21641661 |
| **4** | 1997 | Goa | 72538 |
| **...** | ... | ... | ... |
| **652** | 2019 | Tripura | 914876 |
| **653** | 2019 | Uttar Pradesh | 259987253 |
| **654** | 2019 | Uttarakhand | 9010708 |
| **655** | 2019 | West Bengal | 433058208 |
| **656** | 2020 | Uttarakhand | 10177226 |

657 rows × 3 columns

```
fig=px.scatter(crop_year,x='Crop_Year',y='Production',color='State',title='Year
fig.update_traces(marker_line_width=1,marker_size=10)
fig.update_xaxes(title='Year')
fig.update_layout(title_font_color='Brown')
```

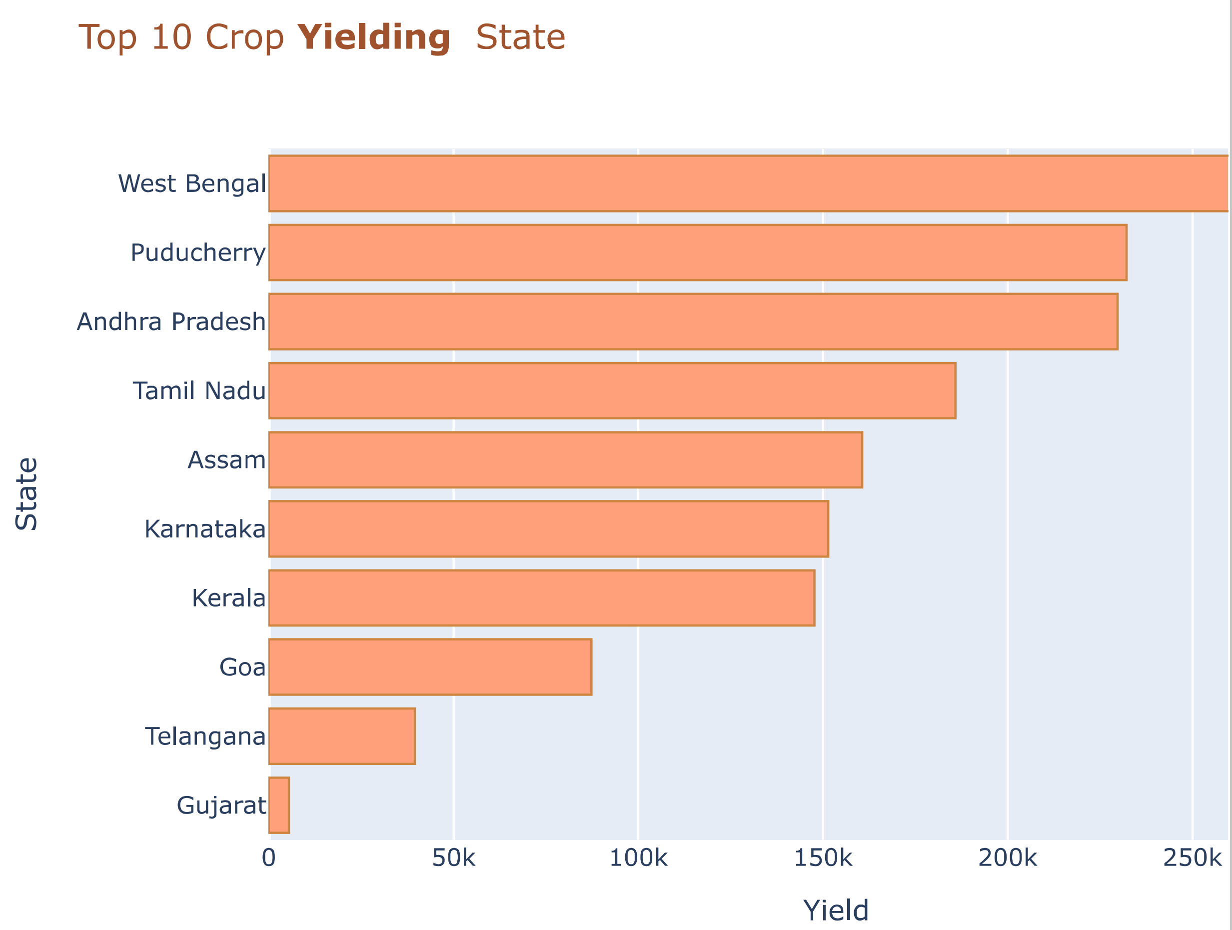



The graph indicates the production of crops in India. Kerala is the highest producing staes in India in all the years with some exception in 2011 and 2015 when Tamil Nadu surpassed Kerala while in 2019 Karnataka was the highest crop producing state followed by Tamil Nadu and Kerala. The graph also shows that before 2002 the production of crop in Tamil Nadu was very low. In 2010 and 2012 also Tamil Nadu registered very low crop production 42.6545M and 40.358M respectively.

```
state_yield=df.groupby('State')[['Yield']].sum()
state_yield=state_yield.sort_values(by=['Yield'],ascending=False).head(10)
state_yield
```

| State | Yield |
|---|---|
| **West Bengal** | 291986.752179 |
| **Puducherry** | 232163.600840 |
| **Andhra Pradesh** | 229735.184723 |
| **Tamil Nadu** | 185813.281648 |
| **Assam** | 160612.648362 |
| **Karnataka** | 151391.867808 |
| **Kerala** | 147710.353263 |
| **Goa** | 87275.955000 |
| **Telangana** | 39508.727733 |
| **Gujarat** | 5471.634563 |

```
fig=px.bar(state_yield,y=state_yield.index,x='Yield',title='Top 10 Crop <b>Yiel
fig.update_yaxes(autorange='reversed')
fig.update_traces(marker_line_width=1,marker_color=' lightsalmon',marker_line_c
fig.update_layout(title_font_color='Sienna')
```



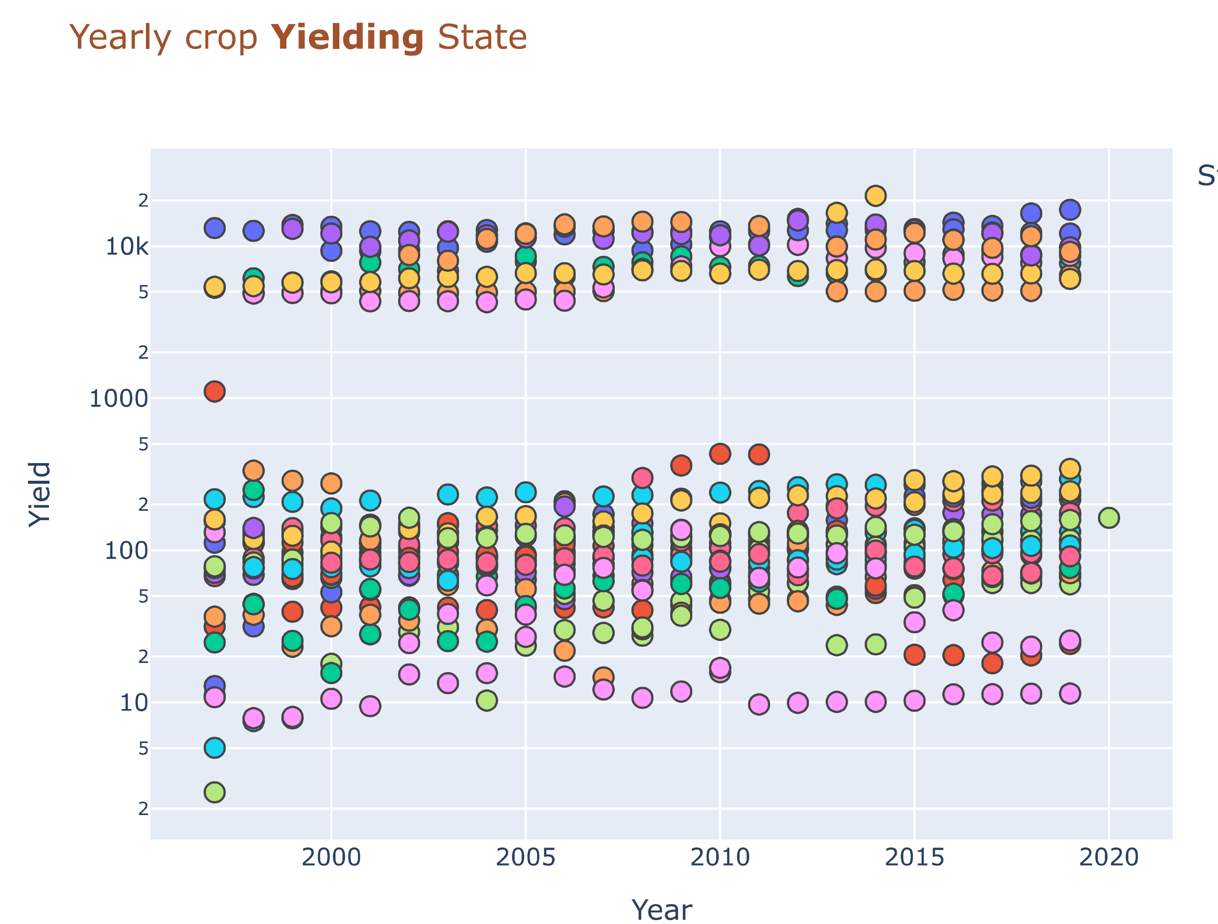

The graph indicates the top 10 crop yielding states in India. West Bengal, Puducherry and Andhra Pradesh are the top three states with highest crop yield, where yield of crops in West Bengal is 291.986k. While in Puducherry and Andhra Pradesh yields are 232.163k and 229.735k followed by Tamil Nadu where yield is 185.813k

```
year_state_production=df.groupby(['Crop_Year','State'])["Yield"].sum()
year_state_production=year_state_production.reset_index()
year_state_production
```

| | Crop_Year | State | Yield |
|---|---|---|---|
| **0** | 1997 | Andhra Pradesh | 112.177745 |
| **1** | 1997 | Arunachal Pradesh | 31.518611 |
| **2** | 1997 | Assam | 5324.162217 |
| **3** | 1997 | Bihar | 154.843858 |
| **4** | 1997 | Goa | 11.075000 |
| **...** | ... | ... | ... |
| **652** | 2019 | Tripura | 107.631607 |
| **653** | 2019 | Uttar Pradesh | 244.026453 |
| **654** | 2019 | Uttarakhand | 158.363811 |
| **655** | 2019 | West Bengal | 12125.197411 |
| **656** | 2020 | Uttarakhand | 163.543553 |

657 rows × 3 columns

```
fig=px.scatter(year_state_production,x='Crop_Year',y='Yield',color='State',titl
fig.update_traces(marker_size=10,marker_line_width=1)
fig.update_xaxes(title='Year')
fig.update_layout(title_font_color='Sienna')
```



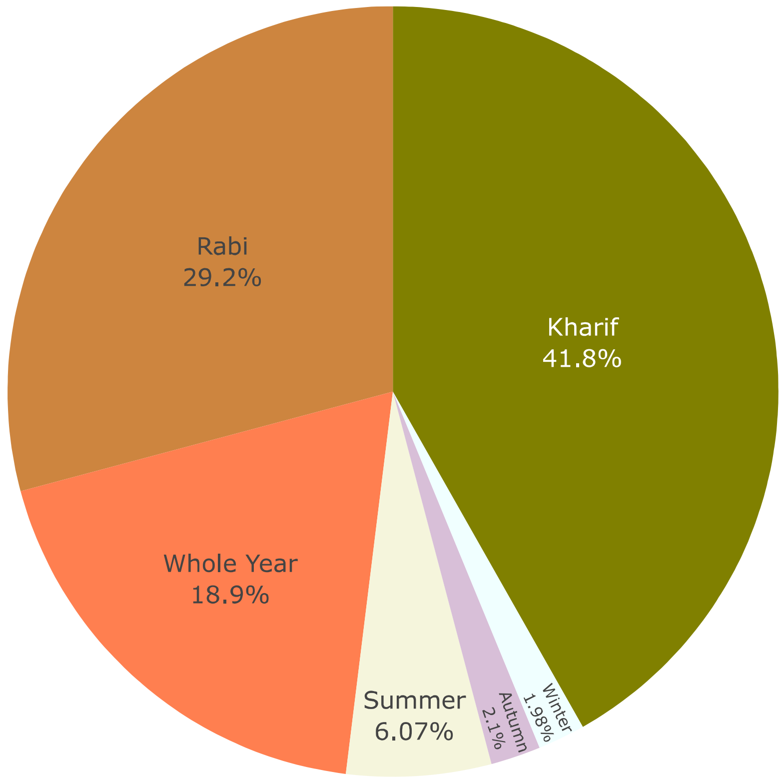

The graph indicates that yield of crop is high in West Bengal, Puducherry, Andhra Pradesh and Tamil Nadu, where the highest yield varies among these states with some exception in 2013 and 2014 when Telangana has the highest crop yield.

```
df['Season'].value_counts()
```

| Season | count |
|---|---|
| **Kharif** | 8232 |
| **Rabi** | 5742 |
| **Whole Year** | 3717 |
| **Summer** | 1195 |
| **Autumn** | 414 |
| **Winter** | 389 |

**dtype:** int64

```
fig = px.pie(values = df['Season'].value_counts(),
             names = ['Kharif', 'Rabi', 'Whole Year', 'Summer', 'Autumn', 'Wint
fig.update_traces(text = df['Season'].value_counts(), textinfo = 'label+percent
                  textposition = 'inside',
                  marker_colors = ['olive', 'peru', 'coral', 'beige', 'thistle'
fig.show()
```



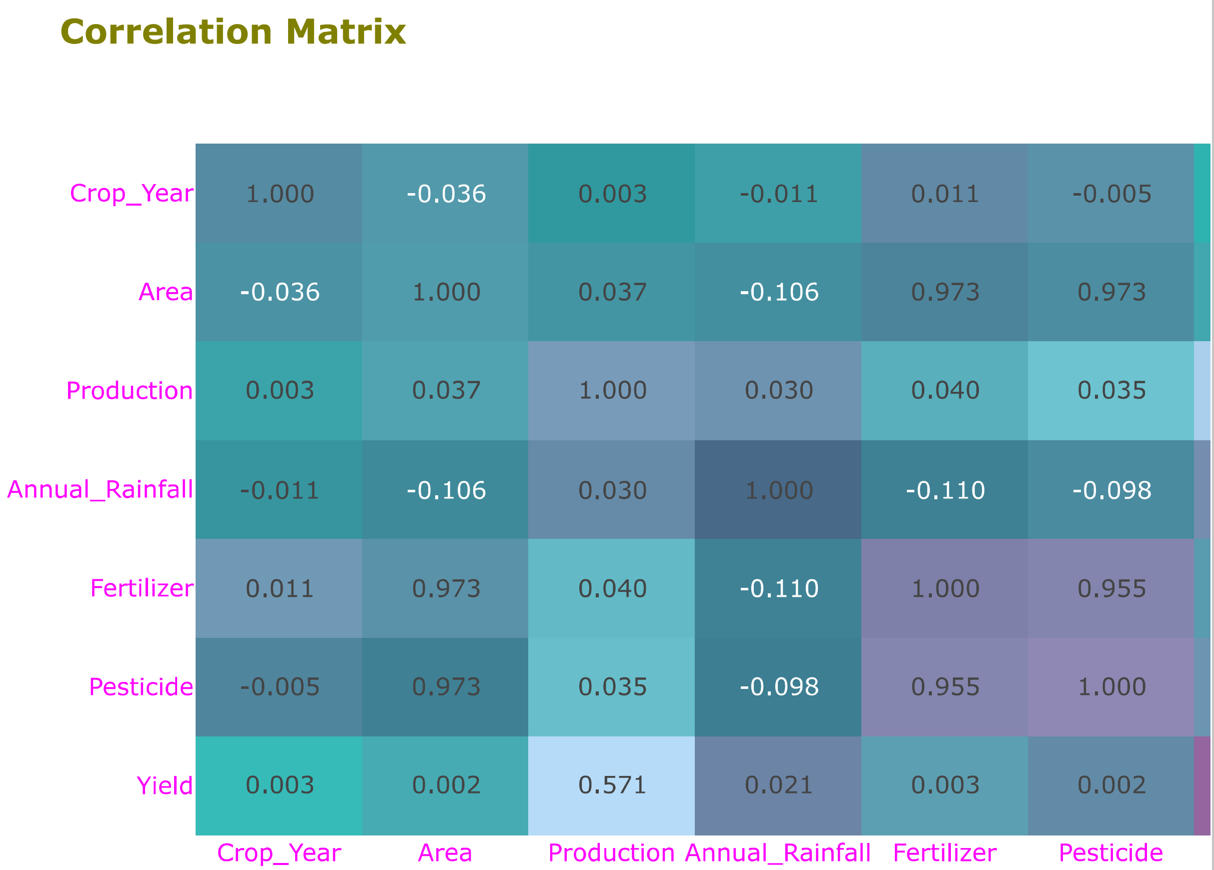

```
fig = px.scatter(df, x = 'Season', y = 'Yield', log_y = True,
                 title = 'Crop Yield In Different <b>Season</b>')
fig.update_traces(marker_color = 'thistle', marker_line_color = 'violet',
                  marker_size = 9, marker_line_width = 1)
fig.update_layout(title_font_color = 'fuchsia')
fig.show()
```


CORRELATION MATRIX

```
correlation=df.corr(numeric_only=True)
correlation
```

| | Crop_Year | Area | Production | Annual_Rainfall | Fertilizer | Pe |
|---|---|---|---|---|---|---|
| **Crop_Year** | 1.000000 | -0.035686 | 0.003366 | -0.011187 | 0.011169 | -( |
| **Area** | -0.035686 | 1.000000 | 0.037441 | -0.106054 | 0.973255 | ( |
| **Production** | 0.003366 | 0.037441 | 1.000000 | 0.029879 | 0.039799 | ( |
| **Annual_Rainfall** | -0.011187 | -0.106054 | 0.029879 | 1.000000 | -0.109734 | -( |
| **Fertilizer** | 0.011169 | 0.973255 | 0.039799 | -0.109734 | 1.000000 | ( |
| **Pesticide** | -0.004657 | 0.973479 | 0.035171 | -0.097657 | 0.954991 | 1 |
| **Yield** | 0.002539 | 0.001858 | 0.570809 | 0.020761 | 0.002862 | ( |

```
fig=px.imshow(correlation,text_auto='.3f',aspect='auto',title='<b>Correlation M
fig.update_layout(title_font_color='olive',font_color='fuchsia')
fig.show()
```




TRAIN_TEST_SPLIT

```
X = df.drop(['Crop', 'Crop_Year', 'Season', 'State', 'Yield'], axis = 1)
y = df['Yield']
```

```
X
```

| | Area | Production | Annual_Rainfall | Fertilizer | Pesticide |
|---|---|---|---|---|---|
| **0** | 73814.0 | 56708 | 2051.4 | 7024878.38 | 22882.34 |
| **1** | 6637.0 | 4685 | 2051.4 | 631643.29 | 2057.47 |
| **2** | 796.0 | 22 | 2051.4 | 75755.32 | 246.76 |
| **3** | 19656.0 | 126905000 | 2051.4 | 1870661.52 | 6093.36 |
| **4** | 1739.0 | 794 | 2051.4 | 165500.63 | 539.09 |
| **...** | ... | ... | ... | ... | ... |
| **19684** | 4000.0 | 2000 | 1498.0 | 395200.00 | 1160.00 |
| **19685** | 1000.0 | 3000 | 1498.0 | 98800.00 | 290.00 |
| **19686** | 310883.0 | 440900 | 1356.2 | 29586735.11 | 96373.73 |
| **19687** | 275746.0 | 5488 | 1356.2 | 26242746.82 | 85481.26 |
| **19688** | 239344.0 | 392160 | 1356.2 | 22778368.48 | 74196.64 |

19689 rows × 5 columns

```
y
```

| | Yield |
|---|---|
| **0** | 0.796087 |
| **1** | 0.710435 |
| **2** | 0.238333 |
| **3** | 5238.051739 |
| **4** | 0.420909 |
| **...** | ... |
| **19684** | 0.500000 |
| **19685** | 3.000000 |
| **19686** | 1.285000 |
| **19687** | 0.016667 |
| **19688** | 1.261818 |

19689 rows × 1 columns

**dtype:** float64

```
from sklearn.model_selection import train_test_split
```

```
x_train,x_test,y_train,y_test=train_test_split(X,y,random_state=42,test_size=0.
```

```
x_train
```

| | Area | Production | Annual_Rainfall | Fertilizer | Pesticide |
|---|---|---|---|---|---|
| **19466** | 357.0 | 1022 | 3340.395455 | 59804.64 | 117.81 |
| **17802** | 75.0 | 196 | 677.500000 | 10005.00 | 12.00 |
| **7014** | 204.0 | 84 | 883.400000 | 21650.52 | 55.08 |
| **14078** | 21054.0 | 24800 | 1287.400000 | 3178311.84 | 6947.82 |
| **8121** | 552393.0 | 379455 | 800.800000 | 52295045.31 | 138098.25 |